

SARGVISION TRAFFIC COPILOT

Vision Document

AI-Powered Traffic Intelligence for Siliguri

Version: 1.0

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Pilot Geography: Siliguri, West Bengal

Primary Stakeholders: Siliguri Police Commissionerate, Traffic Department

Document Purpose: Strategic Vision and Pilot Concept

1. Executive Vision

Siliguri is one of the most strategically important urban and transit centres in North Bengal. Every day, the city handles a complex mix of local commuters, private vehicles, motorcycles, buses, auto-rickshaws, e-rickshaws (Totos), freight vehicles, tourists, pedestrians, and inter-city traffic.

Managing such a dynamic traffic environment is challenging.

Traffic officers often need to make decisions based on multiple fragmented sources of information:

- CCTV monitoring
- Field officers
- Public complaints
- Phone calls
- WhatsApp groups
- Google Maps
- Manual observations
- Traffic control room feeds

The challenge is not simply the lack of information.

The challenge is:

How do we convert large amounts of fragmented traffic information into actionable intelligence?

SARGVISION proposes to address this challenge through:

SARGVISION TRAFFIC COPILOT

An AI-powered traffic intelligence and decision-support platform designed to help traffic authorities understand:

What is happening?
Why is it happening?
Where should attention be prioritized?
What should be investigated next?

The proposed system is not designed to replace traffic officers.

It is designed to give officers an intelligent digital assistant.

2. The Core Problem

Traditional Traffic Monitoring

A typical traffic monitoring environment looks like this:

```
CCTV Screens
  +
Field Officers
  +
Phone Calls
  +
WhatsApp Messages
  +
Google Maps
  +
Manual Reports
  ↓
Human Interpretation
  ↓
Decision
```

The officer is responsible for mentally connecting all these information sources.

This becomes increasingly difficult as:

- the number of roads increases;
- traffic conditions change rapidly;
- multiple incidents occur simultaneously;
- historical context is unavailable;
- recurring patterns are difficult to identify manually.

For example, an officer may know that:

"Sevoke Road is congested."

But several more important questions remain:

- Is this congestion normal for this time of day?
- Is it worse than usual?
- When did it begin?
- Is it spreading to nearby roads?
- Has this happened before?
- Is there a recurring pattern?
- Which locations require immediate attention?
- Which congestion is temporary and which is structural?

Today, answering these questions often requires manual investigation.

3. Our Vision

From Traffic Monitoring to Traffic Intelligence

We want to move from:

SEEING TRAFFIC
↓
to
UNDERSTANDING TRAFFIC
↓
to
PRIORITIZING ATTENTION

The SARGVISION Traffic Copilot acts as an intelligence layer above existing traffic information.

It does not simply display maps.

It continuously analyzes traffic patterns.

It compares:

CURRENT CONDITIONS
VS
WHAT IS NORMALLY EXPECTED

Then it identifies:

- abnormal congestion;
- recurring bottlenecks;
- unusual traffic patterns;
- deteriorating corridors;
- locations requiring investigation.

4. The Simple Concept

Imagine a senior traffic officer starts the day.

Instead of opening multiple systems and manually checking roads, the officer sees:

Good Morning. Here is what requires attention today.

 PRIORITY 1

Sevoke Road

Traffic congestion is currently
48% higher than expected.

The congestion began approximately
35 minutes earlier than usual.

Impact is spreading toward two
connected corridors.

 PRIORITY 2

Hill Cart Road

Travel time reliability has declined
over the past seven days.

Recurring congestion pattern detected
between 5:00 PM and 8:00 PM.

 PRIORITY 3

NJP Corridor

Traffic is building earlier than the
historical evening pattern.

The officer can then ask:

Why is Sevoke Road congested?

The AI investigates the available data and explains what it observes.

This is the central vision.

5. What We Are Building

We are proposing an:

AI-Powered Traffic Operations Intelligence System

The platform will initially consist of five strategic intelligence capabilities.

6. Strategic Product Capability 1

Traffic Situation Intelligence

The Question

What is happening across the city's selected traffic corridors?

The system continuously monitors selected corridors and converts traffic information into a simple operational status.

Example:

SILIGURI TRAFFIC PULSE

Current Status:
MODERATE

Critical Corridors:
3

High Congestion Corridors:
5

Normal Corridors:
7

Most Affected:
Sevoke Road

Current Delay:
+42% compared to expected travel time

The objective is to provide a simplified picture of the city.

7. Strategic Product Capability 2

Traffic Incident Intelligence

The Question

What unusual traffic event is happening right now?

Not every traffic jam is an incident.

Some congestion is expected.

For example:

5:30 PM

Expected Situation:
Moderate congestion

Actual Situation:
Moderate congestion

This is normal.

But consider:

2:15 PM

Expected Travel Time:
12 minutes

Actual Travel Time:
24 minutes

Increase:
100%

This is unusual.

The Traffic Copilot detects this deviation.

It can generate an alert:

Unusual traffic condition detected.

The system can then investigate:

- When did congestion begin?
- Which nearby corridors are affected?
- How unusual is this pattern?
- Has a similar pattern occurred previously?

This does not mean the AI automatically declares:

"There is an accident."

The system should remain evidence-based.

Instead, it can say:

"An unusual traffic anomaly has been detected. Investigation may be required."

This distinction is important.

8. Strategic Product Capability 3

Bottleneck Intelligence

The Question

Where is congestion actually originating?

A common mistake in traffic analysis is to assume:

The place where vehicles are standing is the place causing the problem.

That is not always true.

Consider:

```
ROAD A
  ↓
ROAD B
  ↓
INTERSECTION C
```

↓
BOTTLENECK

Congestion may become visible on Road A.

But the actual cause may be downstream.

For example:

- roadside parking;
- bus stopping;
- road narrowing;
- intersection spillback;
- market activity;
- vehicle merging;
- obstruction.

The initial version of the system will identify:

Probable bottleneck zones.

Example:

CORRIDOR:
Sevoke Road

Recurring Congestion:
5 PM - 8 PM

Probable Bottleneck Zone:
Approximately 300 metres near Location X

Confidence:
High

Pattern:
Recurring across 18 of the last 25 comparable days

This helps authorities know:

Where should we investigate physically?

The AI does not replace field investigation.

It helps prioritize it.

9. Strategic Product Capability 4

Indian Mixed Traffic Intelligence

This is a future capability.

Indian traffic conditions are fundamentally different from highly structured traffic environments.

Siliguri traffic includes:

```
Cars
Motorcycles
Auto-rickshaws
E-rickshaws / Totos
Buses
Trucks
Pedestrians
```

The future platform should use existing CCTV infrastructure and computer vision to understand physical traffic behaviour.

For example:

- Toto clustering;
- illegal parking;
- bus obstruction;
- wrong-way movement;
- pedestrian spillover;
- lane blockage;
- roadside encroachment.

The architecture would be:

```
CCTV
↓
Computer Vision
↓
Vehicle Detection
↓
Vehicle Tracking
↓
Behaviour Detection
↓
Traffic Intelligence
```

This is not required for the initial pilot.

It becomes valuable after the platform establishes trust and gains access to relevant camera feeds.

10. Strategic Product Capability 5

City Mobility Decision Intelligence

This is the long-term vision.

The system will eventually help answer questions such as:

Which corridor should the city prioritize?

Where is infrastructure investment likely to have the highest impact?

Is congestion caused by insufficient road capacity or by downstream bottlenecks?

Which interventions should be tested first?

For example:

Instead of saying:

"This junction looks congested. We should widen it."

The platform may eventually identify:

"Analysis indicates that the majority of recurring delay originates downstream. Junction widening alone may not solve the observed congestion."

This helps decision-makers move toward:

Evidence-based urban mobility planning.

11. The Heart of the Platform

SARGVISION Traffic Operations Copilot

The Copilot is the human interface for the entire intelligence system.

The officer should not need to understand:

- databases;
- analytics engines;
- machine learning models;

- API systems.

The officer should be able to ask questions naturally.

Example Questions

Situation Awareness

What requires attention right now?

Investigation

Why is Sevoke Road congested?

Historical Analysis

Is today's traffic worse than usual?

Comparison

Which corridor has been most unreliable this week?

Pattern Analysis

What time does congestion normally begin on Hill Cart Road?

Daily Operations

Give me today's traffic briefing.

The Copilot retrieves relevant data and uses analytical tools.

It should not invent answers.

12. Why Do We Need This?

Problem 1: Too Much Information

Traffic departments may already have:

- CCTV;
- field personnel;
- digital maps;
- communication systems.

But information is fragmented.

The officer must manually interpret everything.

The Copilot creates:

INFORMATION

↓

ANALYSIS

↓

PRIORITY

↓

ACTION

Problem 2: Lack of Historical Context

A traffic officer may see congestion.

But the important question is:

Is this normal?

The system builds historical intelligence.

Example:

CURRENT TRAVEL TIME

24 minutes

NORMAL FOR THIS TIME

14 minutes

DIFFERENCE

+71%

This provides context.

Problem 3: Difficult to Identify Recurring Problems

Some problems happen:

- every evening;
- every Friday;
- near schools;
- during market hours;
- during festivals.

Humans can observe some patterns.

But continuous data analysis can identify patterns systematically.

Problem 4: Limited Resources

Traffic personnel cannot be everywhere.

Therefore:

Attention should be prioritized.

The system should help answer:

Where should officers focus first?

13. Why Start with a Small Pilot?

Siliguri does not need an expensive Smart City transformation before experimenting with AI.

The recommended approach is:

Start Small. Demonstrate Value. Scale Gradually.

The initial pilot should focus on:

10-15 Strategic Corridors

No New Sensors

No New Hardware

No Signal Controller Integration

Google Maps Data

Historical Traffic Analysis

AI Copilot

This significantly reduces risk.

14. Why This Approach Is Suitable for Siliguri

The proposed approach is specifically designed for a resource-conscious city.

It does not initially require:

- ✗ New Cameras
- ✗ New Sensors
- ✗ Expensive Control Rooms
- ✗ Signal Controller Replacement
- ✗ Large Hardware Procurement
- ✗ City-Wide Deployment

Instead:

EXISTING DIGITAL DATA
+
GOOGLE MAPS CAPABILITIES
+
SARGVISION AI
↓
TRAFFIC INTELLIGENCE

The city can test the value before considering deeper investment.

15. What This Is Not

It is important to clearly define boundaries.

The MVP is not:

- an autonomous traffic control system;
- a replacement for traffic police;
- a replacement for Google Maps;
- a traffic signal controller;
- a surveillance system;
- an automated enforcement system.

It is:

A decision-support and intelligence system.

The final decision remains with human authorities.

16. Relationship with Existing Technologies

The system should complement existing technologies.

For example:

```
GOOGLE MAPS
  ↓
Traffic Movement Information

CCTV
  ↓
Physical Observation

FIELD OFFICERS
  ↓
Ground Reality

SARGVISION
  ↓
Intelligence + Analysis + Copilot
```

The goal is integration of intelligence, not replacement of every existing system.

17. Initial Pilot Vision

We recommend a focused:

SILIGURI TRAFFIC COPILOT PILOT

Duration

Approximately 60–90 days.

Scope

10–15 strategic corridors.

Primary Users

- Traffic Police officers;
- Traffic Control Room personnel;
- Senior traffic leadership.

Initial Outputs

- Daily Traffic Brief;
 - Traffic Priority Alerts;
 - Corridor Performance Analysis;
 - Historical Traffic Patterns;
 - Congestion Anomaly Detection;
 - Bottleneck Investigation;
 - AI Traffic Copilot.
-

18. Example Daily Workflow

Morning

The officer opens the system.

The Copilot generates:

Daily Traffic Brief

Yesterday:

- 3 major traffic anomalies detected
- Highest delay: Corridor A
- Most unreliable corridor: Corridor B

Today:

- High evening congestion risk expected
 - Three locations require monitoring
 - One recurring bottleneck identified
-

Afternoon

The system detects:

Unusual congestion on Corridor X.

The officer asks:

Is this normal?

The system compares:

Current Traffic
vs
Historical Pattern

The response:

No. Current travel time is 62% higher than the expected range for this time and day.

Evening

The officer asks:

Which corridors need additional monitoring?

The system ranks locations based on:

- severity;
 - abnormality;
 - duration;
 - spread.
-

19. The Long-Term Vision

The initial pilot creates the data foundation.

Over time:

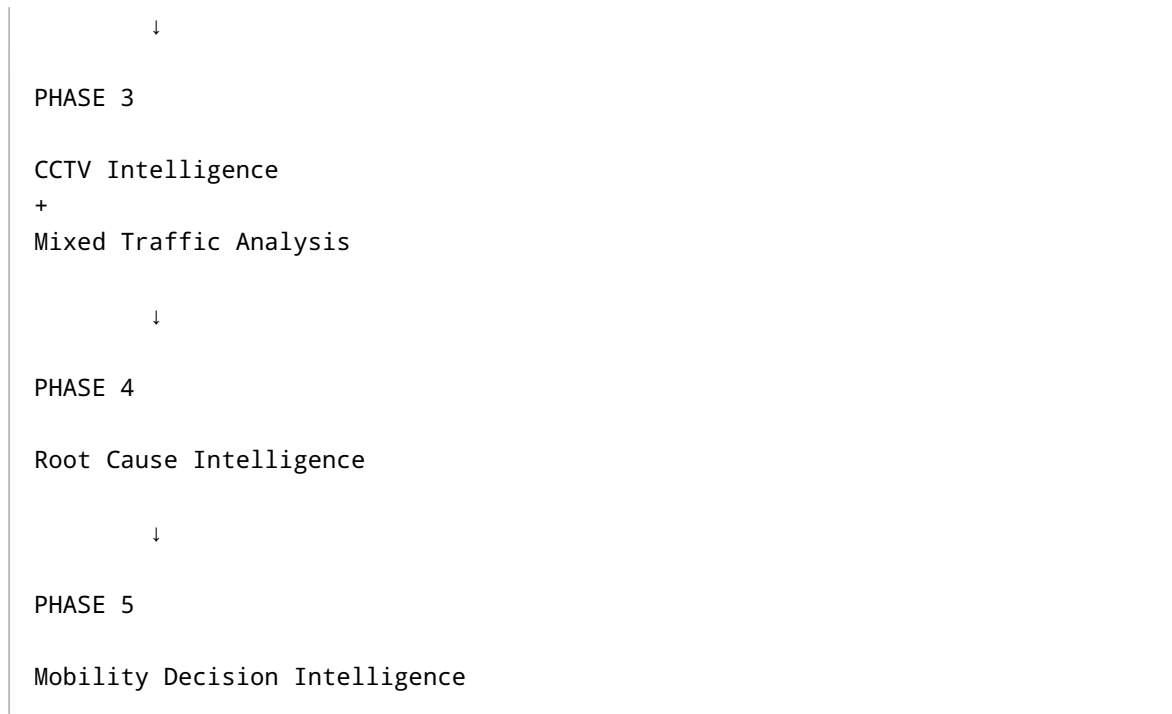
PHASE 1

Traffic Intelligence
+
AI Copilot

↓

PHASE 2

Incident Intelligence
+
Operational Context



Eventually, Siliguri could develop a city-specific mobility intelligence system built around its own:

- traffic patterns;
- road network;
- mobility behaviour;
- operational challenges.

20. Our Strategic Goal

The goal is not simply to create another dashboard.

The goal is to build:

An AI Intelligence Layer for Urban Mobility Operations.

The long-term vision can be summarized simply:

Today, officers monitor traffic.

Tomorrow, AI helps them understand traffic.

Over time, the city develops intelligence that helps it anticipate and manage mobility challenges more effectively.

21. Proposed Pilot Objective

The proposed pilot should answer one fundamental question:

Can AI help Siliguri Traffic Police identify, understand and prioritize traffic problems faster using existing digital data?

If the answer is yes, the system can progressively evolve.

22. Vision Statement

SARGVISION Traffic Copilot aims to make urban traffic operations more intelligent, proactive and evidence-driven by combining traffic data, historical patterns and Agentic AI into a simple decision-support system for human authorities.

23. The Promise

Know what is happening.

Understand why it may be happening.

Prioritize where attention is needed.

Support human officers with intelligence.

SARGVISION TRAFFIC COPILOT

AI-Powered Traffic Intelligence for Siliguri